higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N4

17 August 2026

This marking guideline consists of 7 pages.

tvetpro.co.za



Try our Apps on Play Store

TVET Exam Papers NATED & NCV	**N1-N6** Mathematics NATED	**N1-N4** Engineering Science NATED	**N1-N6** Industrial Electronics NATED	**N4-N6** Electrotechnics NATED	**N4-N6** Mechanotechnics NATED
N5-N6 Power Machines NATED	**N4-N6** Quantity Surveying NATED	**N5-N6** Strength of Materials & Structures NATED	**N1-N3** Engineering Drawing NATED	**N3** Eletro-technology NATED	**N3** Mechano-technology NATED
N4-N6 Building Administration NATED	**N4-N6** Building& Structural Surveying NATED	**N4-N6** Building & Structural Construction NATED	**N1-N3** Building Science NATED	**N1-N3** Building Drawing NATED	**N4** Mechanical Draughting NATED
N1-N3 Electrical Trade Theory NATED	**N2-N3** Diesel Trade Theory NATED	**N5-N6** Strength of Materials & Structures NATED	**N3** Building & Civil Technology NATED	**N1-N2** Fitting and Machining Theory NATED	**N1-N4** Plating and Structural Steel Drawing NATED
N1-N3 Motor Trade Theory NATED	**N5-N6** Mechanical Drawing & Design NATED	**N1-N2** Plumbing Theory NATED	**N4-N6** Fault Finding & Protective Devices NATED	**N2-N3** Instrument Trade Theory NATED	**N5-N6** Engineering Physics NATED
N5-N6 Fluid Mechanics NATED	**N4-N6** Supervisory Management NATED	**N4-N6** Digital Electronics NATED	**N6** Control Systems NATED	**N1-N2** Bricklaying &Plastering Theory NATED	More…

QUESTION 1

1.1
1.1.1 D
1.1.2 C
1.1.3 B
1.1.4 A
1.1.5 B

(5 × 1) (5)

1.2
1.2.1 False
1.2.2 False
1.2.3 False
1.2.4 True
1.2.5 True

(5 × 1) (5)
[10]

QUESTION 2

2.1 2.1.1 $9,4_{16}$ - $1110,11_2$

$9,4_{16} = 1001.0100_2$✓

0*1001,01
1*1110,11

1*1110,11 –2'c – 1*0001,00 +1 = 1*0001,01✓

	0	*	1	0	0	1,	0	1	
	1	*	0	0	0	1,	0	1	
No carry	1	*	1	0	1	0	1	0	✓

-0 1 0 1, 0 1✓ + 1 = $-101,10_2$✓ = $-5,4_8$✓ (6)

2.1.2 $11100,111_2 \times 6,3_8$

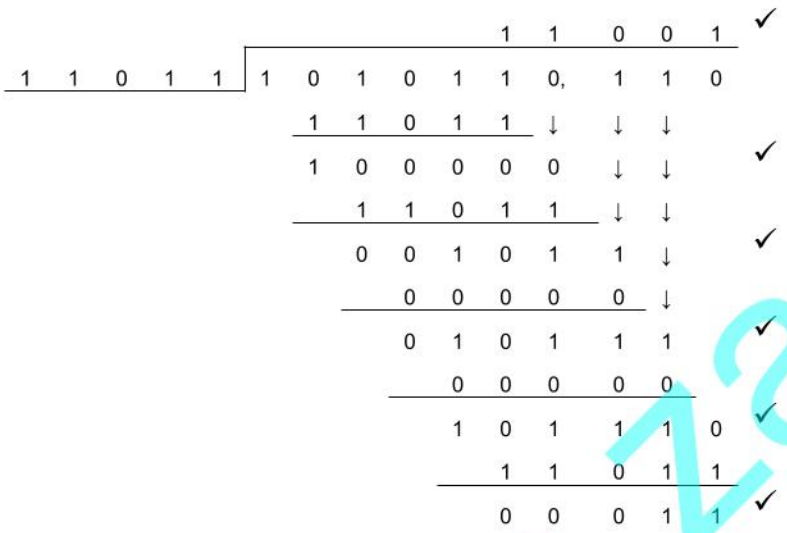
$6,3_8 = 110,011_2$✓

					1	1	1	0	0	1	1	1	
						x	1	1	0	0	1	1	
					1	1	1	0	0	1	1	1	✓
				1	1	1	0	0	1	1	1		
		1	1	1	0	0	1	1	1				✓
	1	1	1	0	0	1	1	1					
1	0	1	1	1	0	0	0	0	0	0	1	0	1 ✓

1 0 1 1 1 0 0 0,0 0 0 1 0 1✓ = $B8,14_{16}$✓ (6)

tvetpro.co.za

2.1.3 $10101,1011_2 \div 110,11_2$



$11,001_2 = 3,125_{10}$ ✓ (7)

2.2 2421; 4221; 5211; 3321; 4311; 642-3; 84-2-1; 631-1; 74-2-1 code.
(Any 3 × 1) (3)

2.3 $110101,11_2$ to 111011,11 Gray✓✓✓ (1 × 3) (3)
[25]

QUESTION 3

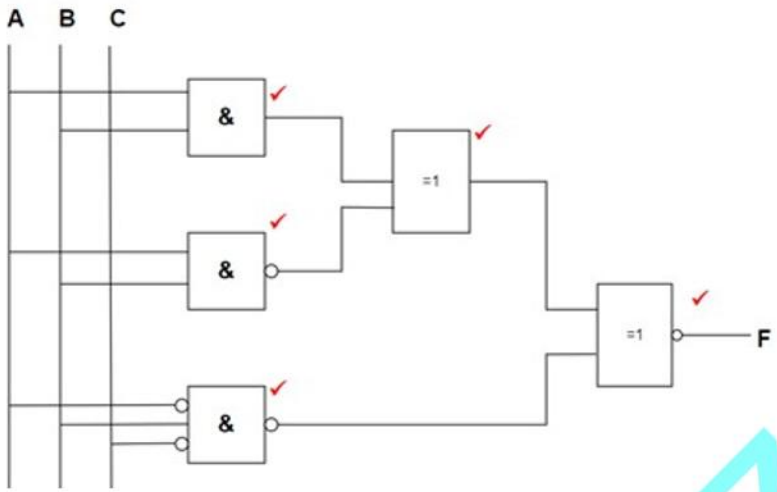
3.1 3.1.1 1
3.1.2 AB + AC
3.1.3 B
3.1.4 AB
(4 × 1) (4)

3.2 3.2.1 $(X+Y)(\overline{X}+Z)(Y+\overline{Z})=F$
$(0+XZ+\overline{X}Y+ZY)(Y+\overline{Z})$✓
$(0+XYZ+\overline{X}Y+ZY+0+0+\overline{X}Y\overline{Z}+0)$✓
$XYZ+\overline{X}Y+ZY+\overline{X}Y\overline{Z}$✓
$Y(XZ+\overline{X}\ \overline{Z})+\overline{X}Y+ZY$
$Y+\overline{X}Y+ZY$✓
$Y(\overline{X}+Z)$✓ (6)

3.2.2 $AB+\overline{\overline{A}+B}+\overline{A\overline{B}C+\overline{A}BC}=F$
$AB+A\overline{B}+(\overline{A}+B+\overline{C})(A+\overline{B}+\overline{C})$✓
$AB+A\overline{B}+(0+\overline{A}\ \overline{B}+\overline{A}\ \overline{C}+AB+0+B\overline{C}+A\overline{C}+\overline{B}\ \overline{C}+\overline{C}$✓
$A(B+\overline{B})+\overline{A}\ \overline{B}+\overline{C}(\overline{A}+A+1)+\overline{C}(B+\overline{B})$✓
$A+\overline{A}\ \overline{B}+\overline{C}$✓
$A+\overline{B}+\overline{C}$✓ (5)

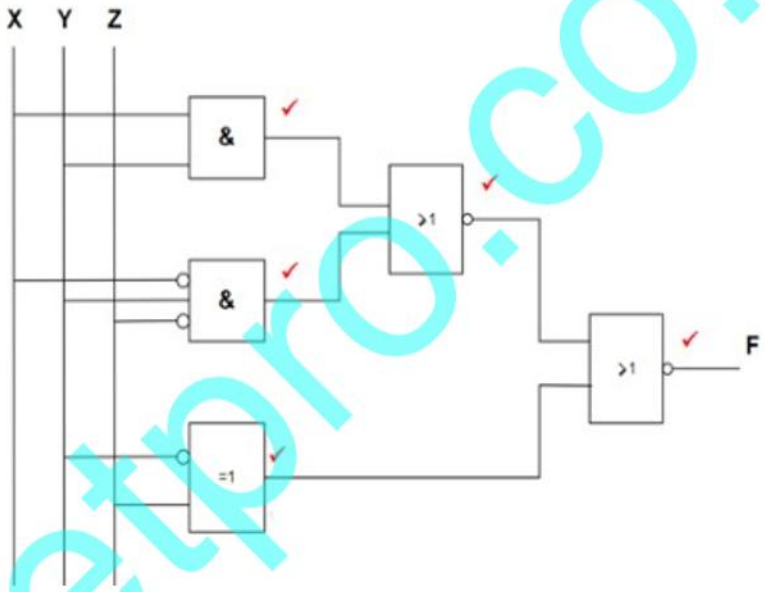
tvetpro.co.za

3.3 3.3.1 $\overline{(AB \oplus \overline{AB})} \oplus \overline{\overline{A}\,B\,\overline{C}} = F$



(5)

3.3.2 $\overline{\overline{XY} + \overline{X}\,Y\,\overline{Z} + \overline{Y} \oplus Z} = F$

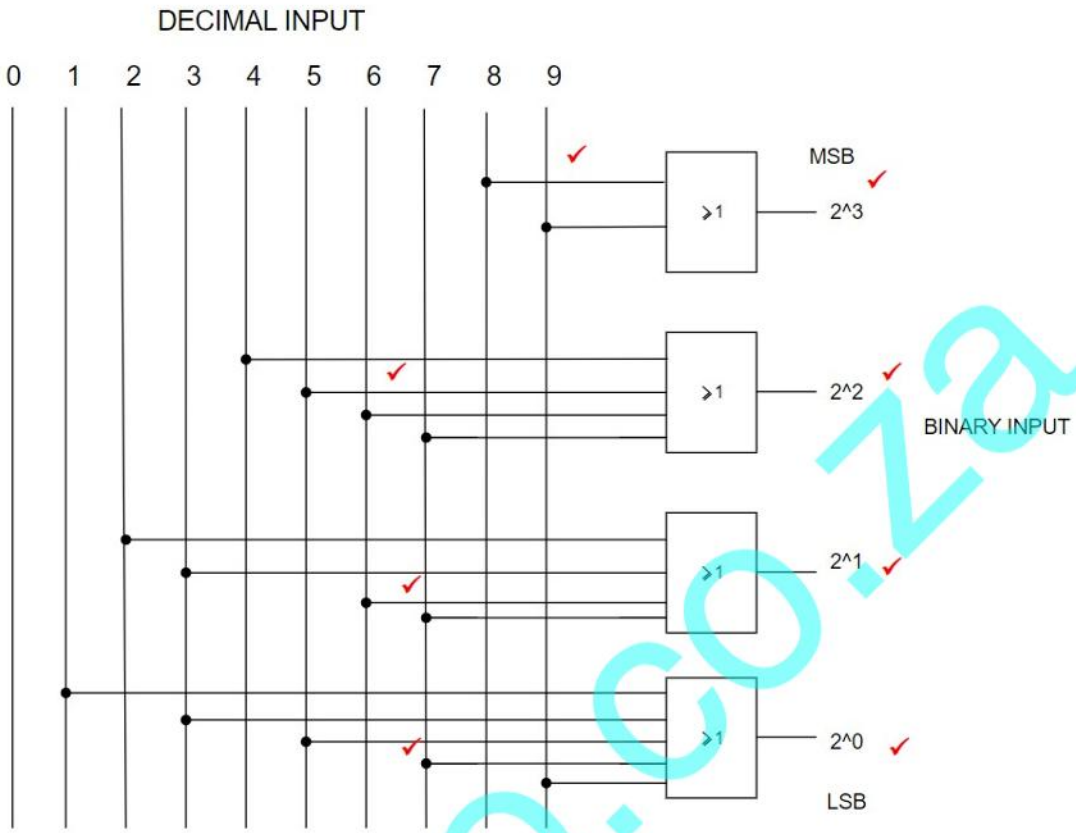


(5)
[25]

QUESTION 4

tvetpro.co.za

4.1 Decimal to binary encoder



		BINARY		
DEC	D	C	B	A
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1

(11)

tvetpro.co.za

4.2



POSITION NO	MASTER	SLAVE
1	Closed	Open
2	Closed	Closed
3	Open	Closed
4	Closed	Closed
5	Closed	Open
6	Closed	Open

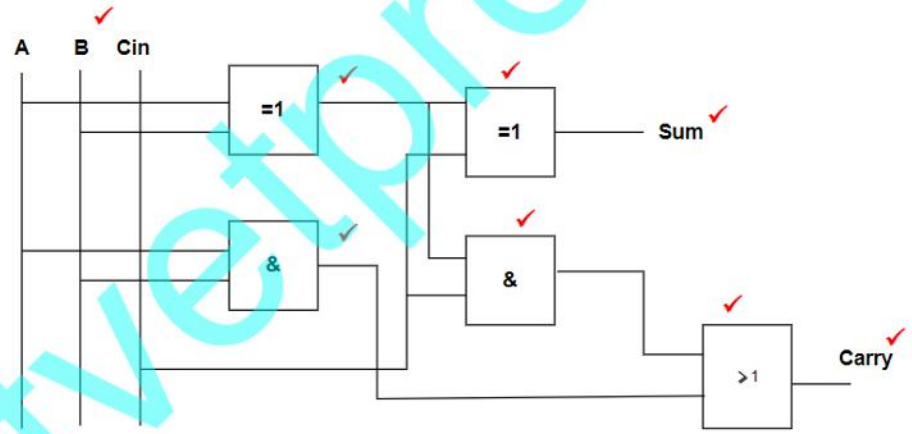
(9)
[20]

QUESTION 5

5.1
- During synchronous switching the flip-flops are clocked simultaneously.✓✓
- In asynchronous switching the output of one flip-flop serves as the clock of the next flip-flop.✓✓

(4)

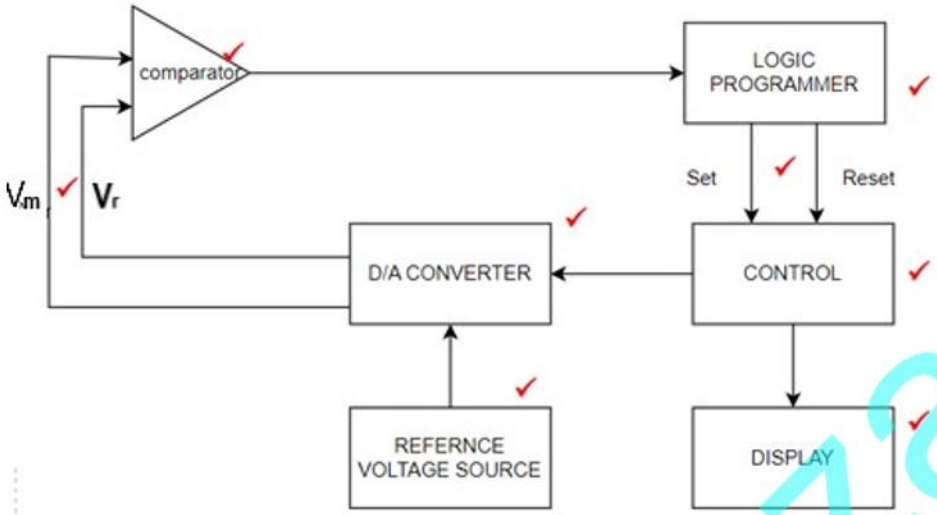
5.2



(8)

tvetpro.co.za

5.3



(8)
[20]

TOTAL: 100

tvetpro.co.za

Copyright reserved